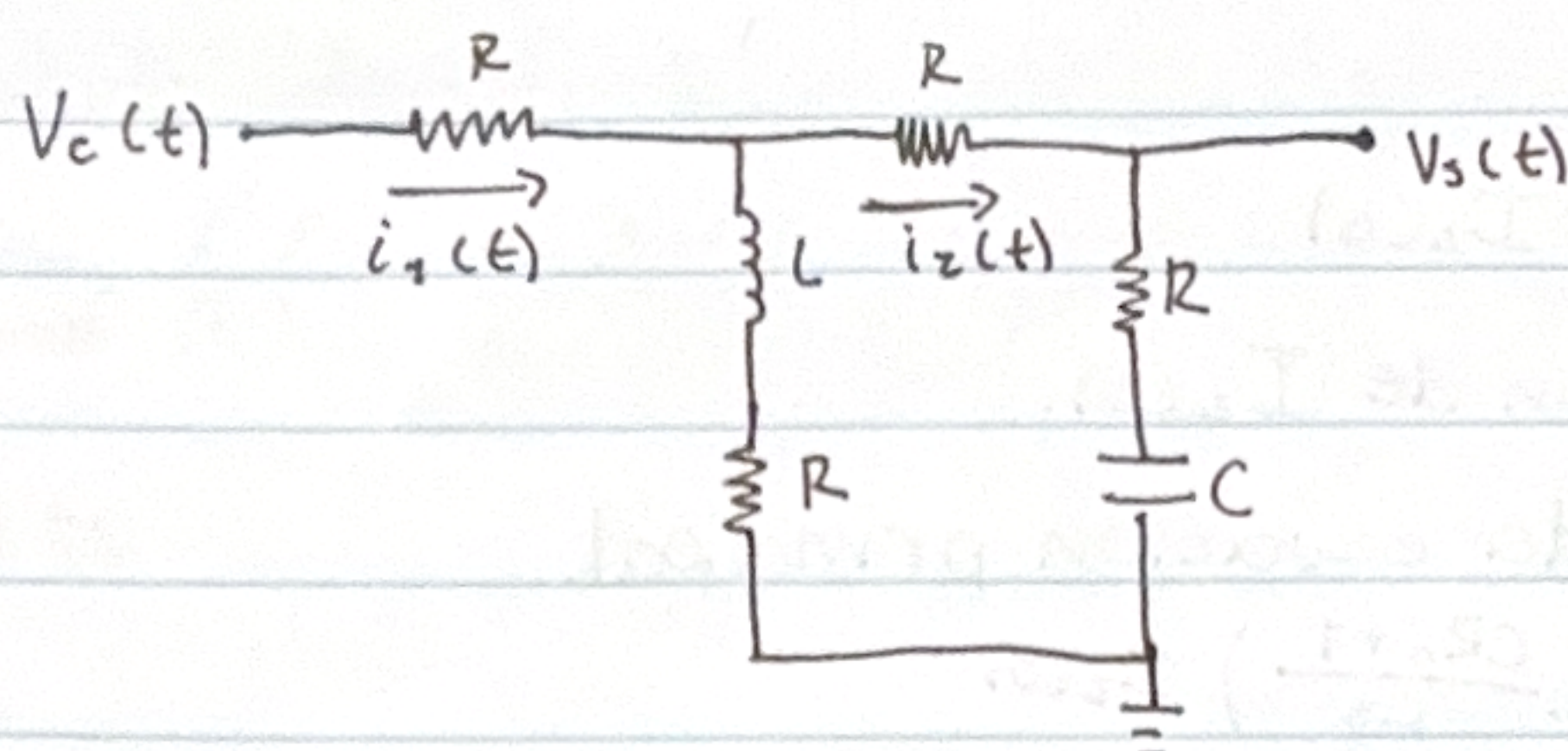


17-09-26



Ecuaciones principales

$$V_c(t) = R i_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} + R [i_1(t) - i_2(t)]$$

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + R [i_1(t) - i_2(t)] = R i_2(t) + R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$V_s(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

Transformada de Laplace

$$V_c(s) = R I_1(s) + L s [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)]$$

$$L s [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)] = R I_2(s) + R I_2(s) + \frac{I_2(s)}{C s}$$

$$V_s(s) = R I_2(s) + \frac{I_2(s)}{C s}$$

Procedimiento algebraico

Función de transferencia

Circuito de
segundo orden

$$\frac{V_s(s)}{V_c(s)} = \frac{(\text{polinomio}) [I_2(s)]}{(2s^2 + 2s + 1) I_2(s)}$$

17-09-26

Procedimiento

1. Agrupar corrientes $I_1(s)$ e $I_2(s)$.
2. Expresar el $V_e(s)$ en función de $I_2(s)$.
3. Despejar $I_1(s)$ de la segunda ecuación principal.

$$V_s(s) = \left(R + \frac{1}{Cs}\right) I_2(s) = \left(\frac{CR_0 + 1}{Cs}\right) I_2(s)$$

$$Ls I_1(s) - Ls I_2(s) + R I_1(s) - R I_2(s) = R I_2(s) + R I_2(s) + \frac{I_2(s)}{Cs}$$

$$Ls I_1(s) + R I_1(s) = R I_2(s) + R I_2(s) + \frac{I_2(s)}{Cs} + Ls I_2(s) + R I_2(s)$$

$$(Ls + R) I_1(s) = \left(Ls + \frac{1}{Cs} + 3R\right) I_2(s)$$

$$I_1(s) = \frac{1}{Ls + R} \left(Ls + \frac{1}{Cs} + 3R\right) I_2(s)$$

$$I_1(s) = \frac{CLs^2 + 3CRs + 1}{CLs^2 + CRs} I_2(s)$$

$$V_e(s) = R I_1(s) + Ls I_1(s) - Ls I_2(s) + R I_1(s) - R I_2(s)$$

$$V_e(s) = (Ls + 2R) I_1(s) - (Ls + R) I_2(s)$$

$$V_e(s) = (Ls + 2R) \frac{CLs^2 + 3CRs + 1}{CLs^2 + CRs} I_2(s) - (Ls + R) I_2(s)$$

$$V_e(s) = \left[\frac{(Ls + 2R)(CLs^2 + 3CRs + 1)}{CLs^2 + CRs} - (Ls + R) \right] I_2(s)$$

$$V_e(s) = \left[\frac{(Ls + 2R)(CLs^2 + 3CRs + 1) - (Ls + R)(CLs^2 + CRs)}{CLs^2 + CRs} \right] I_2(s)$$

$$V_e(s) = \frac{3CLR^2s^2 + 5CR^2s + Ls + 2R}{CLs^2 + CRs}$$

$$\frac{V_e(s)}{V_s(s)} = \frac{3CLR^2s^2 + 5CR^2s + Ls + 2R}{CLs^2 + CRs}$$

17-09-26

Error en estado estacionario se aplica un escalón unitario de entrada

$$e(s) = \lim_{s \rightarrow 0} \frac{1}{s} \left(1 - \frac{CLRs^2 + (CR^2 + L)s + R}{3CLR s^2 + (5CR^2 + L)s + 2R} \right) = \frac{R}{2R}$$

$$e(t) = 0.5V$$

Estabilidad en lazo abierto

Se calculan los polos del sistema (raíces del denominador)

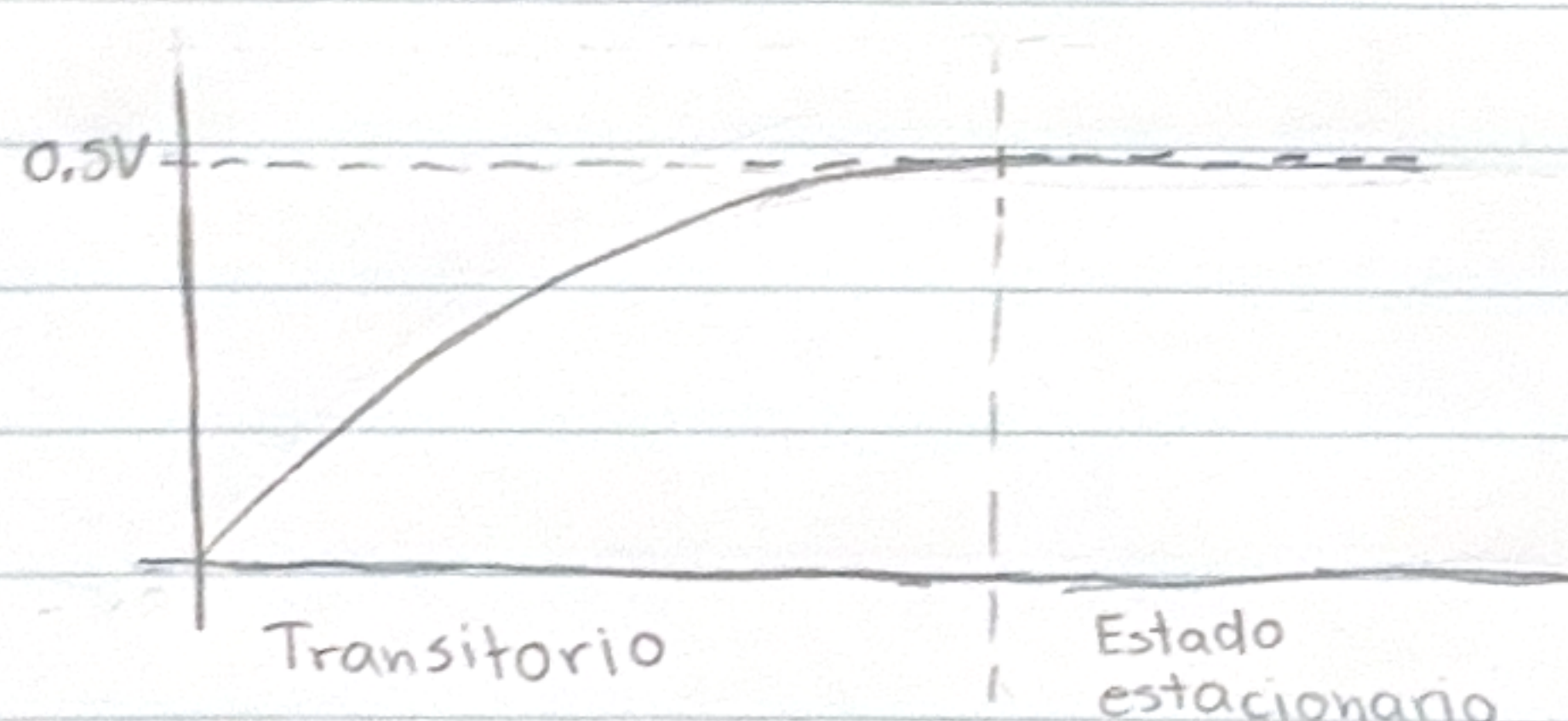
$$R = 10k\Omega \quad \underbrace{3CLR s^2}_a + \underbrace{(5CR^2 + L)s}_b + \underbrace{2R}_c = 0$$

$$C = \cancel{10\mu F} 1\mu F$$

$$L = \cancel{200\mu H} 6.8mH \quad \lambda_1 = -4 \times 10^{-3}, \quad \lambda_2 = -83.33 \times 10^{-6}$$

$$(RC)^2 - 4LC = [(10k\Omega)(10mF)]^2 - 4(200\mu H)(10mF)$$

$$0 \leq 9999.999 \therefore \text{sobreamortiguada}$$



Modelo de ecuaciones integro-diferenciales

A partir de las ecuaciones del sistema, se despejan las variables dependientes, es decir, las corrientes y el voltaje de salida:

$$i_1(t) = \left[V_e(t) + Ri_2(t) - L \frac{d[i_1(t) - i_2(t)]}{dt} \right] \frac{1}{2R}$$

$$i_2(t) = \left[L \frac{d[i_1(t) - i_2(t)]}{dt} + Ri_1(t) - \frac{1}{C} \int i_2(t) dt \right] \frac{1}{3R}$$

$$V_s(t) = Ri_2(t) + \frac{1}{C} \int i_2(t) dt$$